

Unit 1: Introduction to Data Analytics – 2-Mark Questions

1. Define data analytics.

Data analytics is the process of examining raw data using statistical and computational methods to find patterns, draw insights, and support decision-making.

2. What is the importance of data in decision making?

Data helps organizations make evidence-based decisions, minimize risks, identify opportunities, and improve efficiency and performance.

3. List the types of data analytics.

1. Descriptive Analytics
 2. Diagnostic Analytics
 3. Predictive Analytics
 4. Prescriptive Analytics
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4. Differentiate between data analysis and data analytics.

Basis	Data Analysis	Data Analytics
Meaning	It focuses on examining datasets to summarize and interpret information.	It involves using tools, techniques, and models to analyze, predict, and guide decisions.
Purpose	To understand the past and present data trends.	To find patterns, make predictions, and improve future outcomes.
Scope	Narrow – part of the analytics process.	Broad – includes analysis, prediction, and visualization.
Tools Used	Excel, SQL	Python, R, Power BI, Tableau

5. What are the main elements of data analytics?

The main elements are:

- Data Collection
 - Data Cleaning
 - Data Analysis
 - Data Visualization
 - Interpretation and Decision Making
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6. Who is a data analyst?

A data analyst is a professional who collects, organizes, and analyzes data to identify useful insights and help in business decision-making.

7. Who is a data scientist?

A data scientist is an expert who uses programming, statistical methods, and machine learning to build predictive models and extract deep insights from data.

8. Write any two examples of how data analytics is used in real life.

1. In e-commerce, it recommends products to users based on past behavior.
 2. In healthcare, it helps predict disease trends and improve patient care.
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9. What do you mean by descriptive analytics?

Descriptive analytics focuses on summarizing past data to understand what has happened using charts, reports, and dashboards.

10. Define predictive analytics.

Predictive analytics uses statistical models and machine learning techniques to forecast future outcomes based on historical data.

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11. Explain the importance and role of data in modern organizations.

Data plays a crucial role in helping modern organizations make informed and strategic decisions. It serves as the foundation for business intelligence and analytics.

Importance and Role:

- **Informed Decision-Making:** Data helps leaders make evidence-based decisions instead of relying on guesswork.
- **Performance Improvement:** Organizations analyze data to measure efficiency, productivity, and customer satisfaction.
- **Predicting Trends:** Data helps identify future trends in markets and customer behavior.
- **Risk Management:** Analyzing data helps detect fraud, errors, and potential risks early.
- **Innovation:** Insights from data drive innovation in products, services, and marketing strategies.

In short, data helps organizations stay competitive, efficient, and customer-focused in the digital era.

12. Discuss the types of data analytics with suitable examples.

There are **four main types of data analytics**, each serving a different purpose:

Type	Purpose	Example
Descriptive Analytics	Describes what has happened in the past.	A company analyzing monthly sales reports.
Diagnostic Analytics	Explains why something happened.	Investigating why sales dropped in a particular region.
Predictive Analytics	Forecasts what is likely to happen in the future.	Predicting customer churn using machine learning.
Prescriptive Analytics	Suggests actions to achieve desired outcomes.	Recommending best marketing strategies to increase sales.

Together, these analytics types help organizations understand the past, analyze causes, predict the future, and make better decisions.

13. Compare and contrast data analyst and data scientist.

Basis	Data Analyst	Data Scientist
Main Role	Focuses on processing and analyzing existing data to generate reports.	Builds advanced models to predict and interpret complex data patterns.
Skills Required	Strong in Excel, SQL, and data visualization tools.	Strong in programming (Python, R), statistics, and machine learning.
Goal	To find insights and support decisions using structured data.	To create predictive models and derive insights from large, unstructured data.
Tools Used	Excel, Power BI, Tableau	Python, R, TensorFlow, Spark
Output	Dashboards and reports	Predictive models and AI solutions

14. Describe the classification of data analytics.

Data analytics can be classified based on its **purpose and approach** into the following categories:

1. Descriptive Analytics:

- Focuses on summarizing past data to understand what has happened.
- Example: Monthly sales summary reports.

2. Diagnostic Analytics:

- Analyzes data to find out why something happened.
- Example: Identifying reasons for low sales or customer loss.

3. Predictive Analytics:

- Uses statistical models and machine learning to forecast future outcomes.
- Example: Predicting product demand or customer churn.

4. Prescriptive Analytics:

- Recommends actions to achieve desired results or improve performance.
- Example: Suggesting best pricing strategies for profit maximization.

15. Explain the key elements of data analytics with a neat diagram.

The **key elements** (or stages) of data analytics are:

1. **Data Collection:** Gathering relevant data from various sources such as databases, sensors, and surveys.
2. **Data Cleaning:** Removing errors, duplicates, and inconsistencies to improve data quality.
3. **Data Analysis:** Applying statistical or computational techniques to find patterns and relationships.
4. **Data Visualization:** Presenting results using graphs, charts, and dashboards for easy understanding.
5. **Interpretation & Decision-Making:** Drawing conclusions and making business decisions based on the analyzed data.



Unit 2: Python Fundamentals and Statistics – 2-Mark Questions & Answers

16. Why is Python preferred for data analytics?

Python is preferred for data analytics because it is easy to learn, has powerful libraries like NumPy, Pandas, and Matplotlib, and supports data cleaning, analysis, and visualization efficiently.

17. List the features of Python that make it suitable for data science.

1. Simple and readable syntax.
 2. Large collection of data analysis libraries.
 3. Strong community support.
 4. Cross-platform compatibility.
 5. Integration with big data and machine learning tools.
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18. What are the levels of data measurement?

1. **Nominal** – Categorizes data (e.g., gender, color).
 2. **Ordinal** – Ranks data in order (e.g., ratings).
 3. **Interval** – Measures differences but no true zero (e.g., temperature).
 4. **Ratio** – Has a true zero point (e.g., height, weight).
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19. Define mean, median, and mode.

- **Mean:** The average of all data values.
 - **Median:** The middle value when data is arranged in order.
 - **Mode:** The most frequently occurring value in a dataset.
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20. Define standard deviation and variance.

- **Variance:** Measures how far data values spread from the mean.
- **Standard Deviation:** The square root of variance; shows how much data deviates from the mean.

21. What is dispersion in statistics?

Dispersion refers to the extent to which data values are spread out or scattered around the central value (mean or median).

22. Define population and sample.

Term	Definition	Example
Population	The complete set of all items or individuals being studied.	All students in a university.
Sample	A subset selected from the population for analysis.	100 students chosen from the university.

23. What is a confidence interval?

A confidence interval is a range of values that is likely to contain the true population parameter with a specified level of confidence (e.g., 95%).

24. Define sampling error.

Sampling error is the difference between the sample statistic and the actual population parameter caused by observing only a subset of the population.

25. What is the significance of sample mean distribution?

The sample mean distribution helps understand how the mean of different samples relates to the population mean, forming the basis of statistical inference and hypothesis testing.

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26. Explain the importance of Python in data analytics.

Python is one of the most widely used programming languages for data analytics because of its **simplicity**, **powerful libraries**, and **versatility**.

Importance of Python in Data Analytics:

1. Easy to Learn and Use:

Python has simple syntax that makes it beginner-friendly for data professionals.

2. Powerful Libraries:

Libraries like **NumPy**, **Pandas**, **Matplotlib**, and **Seaborn** make data cleaning, analysis, and visualization easy.

3. Supports Machine Learning:

Libraries like **Scikit-learn** and **TensorFlow** allow advanced predictive analytics.

4. High Community Support:

Python has a large active community providing tutorials, tools, and solutions.

5. Integration and Automation:

Python can easily connect with databases, web applications, and other tools, helping automate data workflows.

Conclusion:

Python simplifies every stage of the data analytics process — from collecting data to visualizing and interpreting it.

27. Discuss the levels of data measurement with suitable examples.

Data can be classified into **four levels of measurement**, each defining how data can be analyzed.

Level	Description	Example
Nominal	Data is categorized without any order or ranking.	Gender (Male/Female), Color (Red, Blue)
Ordinal	Data can be ranked but differences between ranks are not measurable.	Movie ratings (Excellent, Good, Poor)
Interval	Ordered data with equal intervals but no true zero.	Temperature (°C, °F)
Ratio	Ordered data with equal intervals and a true zero point.	Height, Weight, Age, Salary

Summary:

Nominal and Ordinal data are **qualitative**, while Interval and Ratio data are **quantitative**.

28. Explain the measures of central tendency and dispersion.

A. Measures of Central Tendency:

These show the center or typical value of a dataset.

1. **Mean:** The average value of all observations.

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

2. **Median:** The middle value when data is arranged in order.
3. **Mode:** The most frequent value in the dataset.

Example:

For data [5, 7, 7, 9, 10]:

Mean = 7.6, Median = 7, Mode = 7.

B. Measures of Dispersion:

These describe how data values are spread around the central value.

1. **Range:** Difference between highest and lowest values.
2. **Variance:** Average of squared differences from the mean.
3. **Standard Deviation:** Square root of variance; measures average deviation from mean.

Example:

For data [4, 6, 8, 10, 12], the range = 8.

Conclusion:

Central tendency shows the “center” of data, while dispersion shows how “spread out” the data is.

29. Describe population, sample, and variance with examples.

Term	Definition	Example
Population	The complete set of all individuals or data points being studied.	All students in a college.
Sample	A small subset selected from the population for analysis.	100 students selected from that college.

Term	Definition	Example
Variance	A measure that shows how much the data values differ from the mean.	If students' marks vary greatly, the variance is high.

Example Explanation:

If population marks are [60, 65, 70, 75, 80],

Mean = 70,

Variance shows how far each mark is from 70 on average.

Conclusion:

A **sample** helps estimate characteristics of the **population**, and **variance** measures the consistency of the data.

30. Explain the concept of confidence interval estimation in statistics.

A **confidence interval (CI)** is a statistical range used to estimate the true value of a population parameter (like mean) based on a sample.

Key Concepts:

1. **Confidence Level:** The probability that the interval contains the true parameter (commonly 95% or 99%).
2. **Margin of Error:** The range above and below the sample mean showing possible variation.
3. **Formula:**

$$\text{Confidence Interval} = \bar{X} \pm Z\left(\frac{\sigma}{\sqrt{n}}\right)$$

where

$\bar{X}$ = sample mean,

Z = z-score for confidence level,

σ = standard deviation,

n = sample size.

Example:

If sample mean = 50, σ = 10, n = 25, and 95% confidence ($Z=1.96$):

$$CI = 50 \pm 1.96 \times (10/\sqrt{25}) = 50 \pm 3.92 \rightarrow (46.08, 53.92)$$

Conclusion:

A 95% confidence interval means we are 95% confident that the true population mean lies between 46.08 and 53.92.

Unit 3: Probability and Types of Testing – 2-Mark Questions & Answers

31. Define probability.

Probability is the measure of how likely an event is to occur.

$$\text{Probability (P)} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

32. What is a probability distribution?

A probability distribution shows how the probabilities are distributed over the possible values of a random variable.

33. List the types of probability distribution.

1. **Discrete Probability Distribution** (e.g., Binomial, Poisson)
 2. **Continuous Probability Distribution** (e.g., Normal, Exponential)
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34. Define hypothesis testing.

Hypothesis testing is a statistical method used to make decisions or inferences about a population parameter using sample data.

35. What are the steps in hypothesis testing?

1. Formulate **null (H_0)** and **alternative (H_1)** hypotheses.
 2. Set the **significance level (α)**.
 3. Choose the **appropriate test** (t-test, ANOVA, etc.).
 4. Calculate the **test statistic**.
 5. **Make a decision** to accept or reject the null hypothesis.
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36. What is a null hypothesis?

A null hypothesis (H_0) is an assumption that there is **no significant difference** or **no effect** between groups or variables.

Example: H_0 : There is no difference between the average marks of two classes.

37. What is an alternative hypothesis?

An alternative hypothesis (H_1) states that there **is a significant difference** or **effect** between groups or variables.

Example: H_1 : There is a difference between the average marks of two classes.

38. What is a t-test used for?

A t-test is used to compare the **means of two groups** to determine if they are significantly different from each other.

Example: Comparing the average performance of two student groups.

39. What is an ANOVA test?

ANOVA (Analysis of Variance) is used to compare the **means of three or more groups** to check if at least one group mean is significantly different.

40. What is a Chi-square test?

A Chi-square test is a statistical test used to examine the **relationship between categorical variables** or to test the **goodness of fit** between observed and expected data.

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41. Explain probability and probability distribution with examples.

1. Probability:

Probability is the measure of how likely an event is to occur.

It ranges from **0 to 1**, where 0 means impossible and 1 means certain.

$$P(E) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$$

Example:

If a coin is tossed, the probability of getting heads = $\frac{1}{2} = 0.5$.

2. Probability Distribution:

A probability distribution shows how probabilities are assigned to each possible value of a random variable.

Types:

- **Discrete Distribution:** Deals with countable outcomes.
Example: Binomial distribution (e.g., number of heads in coin tosses).
- **Continuous Distribution:** Deals with measurable outcomes.
Example: Normal distribution (e.g., human height or weight).

Conclusion:

Probability tells how likely an event is, while a probability distribution explains how probabilities are spread across all outcomes.

42. Discuss sampling and sampling distribution in detail.

1. Sampling:

Sampling is the process of selecting a small group (sample) from a large population to draw conclusions about the entire population.

Types of Sampling:

- **Random Sampling:** Every item has an equal chance of selection.
- **Systematic Sampling:** Every n th item is selected.
- **Stratified Sampling:** Population divided into groups (strata) and samples taken from each.
- **Cluster Sampling:** Population divided into clusters; entire clusters are randomly selected.

2. Sampling Distribution:

A sampling distribution is the probability distribution of a statistic (like mean or proportion) obtained from multiple samples of the same size from the population.

Example:

If we repeatedly take samples of 30 students and calculate their mean height, the distribution of those means forms a *sampling distribution*.

Conclusion:

Sampling saves time and cost, while sampling distribution helps in making predictions about population parameters.

43. Explain the concept and steps of hypothesis testing with an example.

Concept:

Hypothesis testing is a procedure to decide whether a statement about a population is true or false using sample data.

Steps:

1. State the Hypotheses:

- Null Hypothesis (H_0): No difference or effect exists.
- Alternative Hypothesis (H_1): A difference or effect exists.

2. Set the Significance Level (α): Commonly 0.05 (5%).

3. Choose the Test Statistic: (t-test, ANOVA, etc.)

4. Calculate the Test Value: Using sample data.

5. Make a Decision:

- If $p\text{-value} < \alpha$, reject H_0 .
- If $p\text{-value} \geq \alpha$, accept H_0 .

Example:

To test whether the average height of students = 160 cm:

- $H_0: \mu = 160$
- $H_1: \mu \neq 160$

After performing the t-test, if $p < 0.05 \rightarrow$ reject H_0 .

Conclusion:

Hypothesis testing helps make data-driven decisions instead of assumptions.

44. Compare t-test, ANOVA test, and Chi-square test.

Basis	t-test	ANOVA Test	Chi-square Test
Purpose	Compares means of two groups	Compares means of three or more groups	Tests association between categorical variables
Type of Data	Continuous	Continuous	Categorical
Hypothesis	H_0 : No difference between two means	H_0 : All group means are equal	H_0 : No relationship between variables
Statistic Used	t-statistic	F-statistic	χ^2 (Chi-square) statistic
Example	Compare average marks of boys and girls	Compare sales of 3 regions	Test link between gender and product choice

Conclusion:

All three tests are used in hypothesis testing but differ in the **type of data** and **number of groups** being compared.

45. Explain one-way ANOVA with its assumptions and interpretation.

1. Definition:

One-way ANOVA (Analysis of Variance) is used to test whether the means of **three or more independent groups** are significantly different.

2. Example:

Comparing the average test scores of students from three different classes.

3. Assumptions of One-way ANOVA:

1. Data is **normally distributed**.
2. Samples are **independent** of each other.
3. All groups have **equal variances** (homogeneity of variance).
4. The variable is **quantitative**.

4. Steps:

- Formulate hypotheses:
 H_0 : All group means are equal.
 H_1 : At least one group mean is different.
- Calculate **F-statistic** using between-group and within-group variances.
- Compare the calculated F with the critical F-value.

5. Interpretation:

- If $p < 0.05$, reject $H_0 \rightarrow$ At least one mean is different.
- If $p \geq 0.05$, fail to reject $H_0 \rightarrow$ No significant difference.

Conclusion:

One-way ANOVA helps determine if there are meaningful differences among group means, often used in business, education, and scientific studies.

Unit 4: Regression, Classification, and Clustering – 2-Mark Questions & Answers

46. What is regression analysis?

Regression analysis is a statistical method used to study the relationship between a **dependent variable** and one or more **independent variables** to predict outcomes.

Example: Predicting sales based on advertising spend.

47. Differentiate between linear and logistic regression.

Basis	Linear Regression	Logistic Regression
Purpose	Predicts a continuous value.	Predicts a categorical (yes/no) outcome.
Output	Gives numerical results.	Gives probability values between 0 and 1.
Equation Type	Linear equation ($y = mx + c$).	Sigmoid (S-shaped) function.
Example	Predicting house price.	Predicting whether a student passes or fails.

48. Define independent and dependent variables.

- **Independent Variable:** The input or predictor variable that influences the outcome.
- **Dependent Variable:** The output or variable being predicted or measured.

Example:

In predicting sales from advertising,

Advertising = Independent variable, Sales = Dependent variable.

49. What is clustering?

Clustering is a technique in unsupervised learning that groups similar data points into clusters so that objects in the same group are more similar to each other than to those in other groups.

50. What is K-Means clustering?

K-Means is a clustering algorithm that divides data into **K clusters** based on similarity by minimizing the distance between data points and their cluster center (centroid).

51. What is hierarchical clustering?

Hierarchical clustering builds a tree-like structure of clusters (called a **dendrogram**) by either merging smaller clusters (agglomerative) or splitting larger ones (divisive).

52. What is a decision tree?

A decision tree is a classification and prediction model that splits data into branches based on feature values, forming a tree-like structure with decision nodes and leaf nodes.

53. Define confusion matrix.

A confusion matrix is a table used to evaluate the performance of a classification model by comparing **actual** and **predicted** values.

It includes:

- True Positives (TP)
 - True Negatives (TN)
 - False Positives (FP)
 - False Negatives (FN)
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54. What is supervised learning?

Supervised learning is a type of machine learning where models are trained using **labeled data** — that is, data with known input-output pairs.

Example: Predicting exam results based on past performance.

55. What is unsupervised learning?

Unsupervised learning is a machine learning method where models learn patterns from **unlabeled data**, without predefined outputs.

Example: Grouping customers based on purchase behavior using clustering.

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56. Explain linear regression with an example and equation.

Definition:

Linear regression is a statistical technique used to model the relationship between a dependent variable (Y) and one or more independent variables (X) using a straight-line equation.

Equation:

$$Y = b_0 + b_1X + e$$

Where:

- Y = Dependent variable
- X = Independent variable
- b_0 = Intercept
- b_1 = Slope (change in Y for one unit change in X)
- e = Error term

Example:

Predicting house prices based on square footage:

$$\text{Price} = 50,000 + 100 \times \text{Area}$$

Interpretation:

If the area increases by 1 sq. ft., the price increases by ₹100.

Conclusion:

Linear regression helps in prediction when the relationship between variables is approximately linear.

57. Describe logistic regression and its applications.

Definition:

Logistic regression is used when the **dependent variable is categorical** (e.g., Yes/No, 0/1). It predicts the probability that an observation belongs to a particular category.

Equation:

$$P(Y = 1) = \frac{1}{1 + e^{-(b_0 + b_1 X)}}$$

Applications:

1. **Spam detection:** Classifying emails as spam or not spam.
2. **Medical diagnosis:** Predicting whether a patient has a disease (Yes/No).
3. **Customer churn:** Predicting whether a customer will leave or stay.
4. **Loan approval:** Approving or rejecting applications.

Conclusion:

Logistic regression is widely used for binary classification and decision-making tasks.

58. Explain the steps involved in K-Means clustering.

Definition:

K-Means is an unsupervised learning algorithm used to divide data into *K clusters* based on similarity.

Steps:

1. **Select K:** Choose the number of clusters (K).
2. **Initialize Centroids:** Randomly select K data points as initial cluster centers.
3. **Assign Points:** Assign each data point to the nearest centroid based on distance (e.g., Euclidean distance).
4. **Update Centroids:** Calculate new centroids as the average of all points in a cluster.
5. **Repeat:** Repeat steps 3–4 until centroids no longer change significantly.

Example:

Grouping customers into 3 clusters based on income and spending score.

Conclusion:

K-Means helps find natural groupings in data for analysis and segmentation.

59. Differentiate between K-Means and Hierarchical clustering.

Basis	K-Means Clustering	Hierarchical Clustering
Approach	Partitions data into K clusters directly.	Builds a hierarchy of clusters (tree structure).
Number of Clusters	Must be specified in advance.	Not required; can be chosen from dendrogram.
Speed	Faster for large datasets.	Slower for large datasets.
Output	Flat clusters.	Hierarchical (nested) clusters.
Example	Customer segmentation by K value.	Dendrogram showing relationships among customers.

Conclusion:

K-Means is preferred for large datasets, while Hierarchical clustering is better for smaller datasets where relationships need visualization.

60. Explain decision tree algorithm and its working.

Definition:

A decision tree is a classification model that splits data into branches based on feature values to reach a decision.

Working Steps:

1. **Select Attribute:** Choose the feature that best splits the data using measures like *Information Gain* or *Gini Index*.
2. **Split Data:** Divide the dataset into subsets based on the selected attribute.
3. **Repeat Process:** Continue splitting until all data points belong to pure classes or stopping criteria are met.
4. **Prediction:** For a new input, follow the branches of the tree to reach a leaf node (output).

Example:

A decision tree predicting whether a person buys a car based on income and age.

Conclusion:

Decision trees are simple to interpret and visualize but may overfit if not pruned properly.

61. Describe the components and interpretation of a confusion matrix.

Definition:

A confusion matrix is used to evaluate the performance of a classification model by comparing actual vs. predicted outcomes.

Actual / Predicted **Predicted Positive** **Predicted Negative**

Actual Positive True Positive (TP) False Negative (FN)

Actual Negative False Positive (FP) True Negative (TN)

Metrics Derived:

- **Accuracy:** $(TP + TN) / \text{Total}$
- **Precision:** $TP / (TP + FP)$
- **Recall:** $TP / (TP + FN)$
- **F1 Score:** Harmonic mean of precision and recall

Interpretation:

The confusion matrix helps understand not only accuracy but also types of errors a model makes.

62. Compare regression, classification, and clustering approaches.

Basis	Regression	Classification	Clustering
Purpose	Predicts continuous numeric values.	Predicts categorical class labels.	Groups similar data points without labels.
Learning Type	Supervised	Supervised	Unsupervised
Output	Numeric (e.g., sales, marks).	Discrete classes (e.g., pass/fail).	Cluster labels (e.g., customer groups).
Example	Predicting house prices.	Email spam detection.	Customer segmentation.
Algorithms	Linear Regression	Decision Tree, Logistic Regression	K-Means, Hierarchical

63. Discuss the importance of model evaluation metrics in classification.

Importance:

1. **Accuracy:** Measures how often predictions are correct.
2. **Precision:** Indicates how many predicted positives are actually correct.
3. **Recall (Sensitivity):** Measures how many actual positives are correctly identified.
4. **F1 Score:** Balances precision and recall — useful for imbalanced datasets.
5. **ROC Curve & AUC:** Show the trade-off between true positive rate and false positive rate.

Conclusion:

Evaluation metrics ensure that classification models are not only accurate but also reliable and fair across different classes.

64. Explain how clustering helps in customer segmentation.**Definition:**

Customer segmentation is the process of dividing customers into groups based on similar behavior or characteristics.

Clustering Role:

1. **Data Grouping:** Groups customers by purchase habits, spending, or preferences.
2. **Personalization:** Helps businesses target each group with customized offers.
3. **Trend Identification:** Identifies patterns in customer data for marketing strategies.
4. **Profit Maximization:** Improves product recommendations and customer retention.

Example:

A retailer uses clustering to separate customers into *low-, medium-, and high-spending* groups.

Conclusion:

Clustering helps businesses understand their customers better, leading to improved satisfaction and profitability.

65. Write short notes on supervised and unsupervised learning with examples.

Aspect	Supervised Learning	Unsupervised Learning
Definition	Learning from labeled data with known outputs.	Learning from unlabeled data to find patterns.
Goal	Predict outcomes or classify data.	Discover hidden patterns or groups.
Algorithms	Linear/Logistic Regression, Decision Trees.	K-Means, Hierarchical Clustering.
Example	Predicting exam results using past scores.	Grouping customers by purchasing behavior.

Conclusion:

Supervised learning deals with prediction and classification, while unsupervised learning focuses on pattern discovery and data grouping.

Unit 5: Data Visualization Using Power BI — 2-Mark Questions & Answers

66. What is data visualization?

Answer:

Data visualization is the process of representing data in graphical or pictorial form, such as charts, graphs, and dashboards, to make it easier to understand and interpret patterns and trends.

67. What is Power BI?

Answer:

Power BI is a **business analytics tool** developed by Microsoft that allows users to connect, visualize, and share data insights through interactive reports and dashboards.

68. List any four features of Power BI.

Answer:

1. Data connectivity with multiple sources
 2. Interactive dashboards and reports
 3. Real-time data monitoring
 4. Data modeling and transformation
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69. Define data transformation.

Answer:

Data transformation is the process of **cleaning, shaping, and modifying raw data** into a suitable format for analysis or visualization.

70. What is a data model in Power BI?

Answer:

A data model in Power BI is a **structured representation of data tables** and their relationships used for analysis and reporting.

71. What are the types of visualizations available in Power BI?

Answer:

Common Power BI visualizations include:

1. Bar chart
 2. Pie chart
 3. Line chart
 4. Map visualization
 5. Table and matrix
 6. Card and KPI visuals
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72. What are dashboards and reports in Power BI?

Feature	Dashboard	Report
Definition	A single-page summary view of visuals.	A multi-page detailed view with visuals.
Purpose	Provides at-a-glance insights.	Provides in-depth analysis.
Interactivity	Limited interaction.	Highly interactive.
Data Source	Can combine visuals from multiple reports.	Based on a single dataset.

73. How can data be imported from different sources in Power BI?**Answer:**

Data can be imported using “**Get Data**” option from various sources like:

- Excel files
 - SQL Server
 - Web data
 - Cloud services (Azure, Google Drive)
 - CSV files
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74. What is Power Query Editor used for?

Answer:

Power Query Editor is used for **data cleaning, transformation, and preparation** before loading it into Power BI for analysis and visualization.

75. What is meant by publishing a Power BI report?

Answer:

Publishing a Power BI report means **uploading the report from Power BI Desktop to the Power BI Service** so it can be **shared online** with other users or teams.

5-Mark Questions & Answers

76. Explain the importance of visualization and analytic tools in data analytics.

Answer:

Visualization and analytics tools play a vital role in understanding complex data.

Importance:

1. **Simplifies Data Understanding:** Converts raw data into clear visuals.
2. **Faster Decision Making:** Enables quick analysis through interactive charts.
3. **Identifies Patterns and Trends:** Helps detect hidden insights.
4. **Enhances Communication:** Makes data insights easy to explain to non-technical users.
5. **Improves Accuracy:** Reduces human error by automating analysis.

Conclusion:

Visualization tools like Power BI, Tableau, and Excel turn data into actionable insights for better decision-making.

77. Discuss the architecture and components of Power BI.

Answer:

Power BI architecture consists of several components working together for data analysis and reporting.

Main Components:

1. **Power BI Desktop:** Used for data import, transformation, and report creation.
2. **Power BI Service:** Cloud-based service for publishing and sharing reports.
3. **Power BI Data Gateway:** Connects on-premises data sources with the cloud.
4. **Power BI Mobile App:** Allows users to access reports on mobile devices.
5. **Power BI Report Server:** Used for hosting reports on local (on-premises) servers.
6. **Power Query:** Used for cleaning and transforming data.

Conclusion:

The architecture ensures smooth data flow from sources to visualization and sharing.

78. Describe the process of getting data from different sources in Power BI.

Steps:

1. **Open Power BI Desktop.**
2. Click **“Get Data”** on the Home tab.
3. Choose a data source (Excel, SQL, Web, etc.).
4. **Connect** to the source and select tables.
5. Use **Power Query Editor** to clean or transform data.
6. **Load** the data into Power BI for visualization.

Example:

Connecting Power BI to an Excel file containing sales data and creating charts from it.

Conclusion:

Power BI supports multiple data sources, making it flexible and easy for data integration.

79. Explain data transformation and data modeling in Power BI.

Aspect	Data Transformation	Data Modeling
Definition	Process of cleaning and preparing data.	Structuring data for analysis and relationships.
Tool Used	Power Query Editor	Model view in Power BI

Aspect	Data Transformation	Data Modeling
Purpose	Remove duplicates, rename columns, filter rows, etc.	Define relationships between tables and create measures.
Example	Changing data types or merging tables.	Relating “Sales” and “Customer” tables by CustomerID.

Conclusion:

Both transformation and modeling are essential for accurate and efficient analysis in Power BI.

80. List and explain various types of visualizations in Power BI with examples.

Common Visualizations in Power BI:

Type	Description	Example
Bar/Column Chart	Shows comparison among categories.	Sales by region.
Pie/Donut Chart	Displays proportion of total.	Market share by brand.
Line Chart	Shows trends over time.	Monthly revenue growth.
Map Visualization	Displays geographical data.	Sales by country.
Table/Matrix	Shows detailed data with rows and columns.	Product-wise sales list.
Card/KPI Visual	Displays key metrics.	Total profit or growth %.

Conclusion:

Power BI offers a variety of visuals that help users analyze data effectively from different perspectives.

81. Explain the steps involved in creating and publishing reports in Power BI.

Steps:

1. **Get Data:** Import data from multiple sources.
2. **Transform Data:** Clean and shape data using Power Query.

3. **Model Data:** Define relationships between tables.
4. **Create Visuals:** Use charts, tables, and cards to create reports.
5. **Format and Design:** Add titles, filters, and slicers for interactivity.
6. **Publish Report:** Click **Publish** to upload it to the Power BI Service.
7. **Share:** Distribute reports with team members or management.

Conclusion:

The process allows users to move seamlessly from raw data to professional, shareable insights.

82. Discuss how dashboards and analytical reports are created using Power BI.

Steps to Create Dashboards and Reports:

1. **Prepare Data:** Connect and clean data.
2. **Build Reports:** Create visuals and charts in Power BI Desktop.
3. **Publish Reports:** Upload them to Power BI Service.
4. **Create Dashboard:** Pin visuals from different reports to a single page.
5. **Add Filters:** Use slicers for interactive analysis.
6. **Share:** Share dashboards with users or embed them in apps.

Conclusion:

Dashboards provide a summary of key performance metrics, while reports offer detailed data insights.

83. Explain the use cases of Power BI dashboards in industry.

Common Use Cases:

1. **Sales Analysis:** Track regional or product-wise sales performance.
2. **Financial Reporting:** Monitor revenue, profit, and expenses.
3. **Human Resources:** Analyze employee performance and attendance.
4. **Marketing Analytics:** Evaluate campaign success and ROI.
5. **Healthcare:** Monitor patient data and hospital performance.
6. **Education:** Track student progress and institutional metrics.

Conclusion:

Power BI dashboards are widely used across industries for real-time performance monitoring and data-driven decisions.

84. Discuss the role of Power BI in business decision making.

Role of Power BI:

- 1. **Data Integration:** Combines multiple data sources into one platform.
- 2. **Interactive Visualization:** Helps understand business performance visually.
- 3. **Real-time Insights:** Enables timely business decisions.
- 4. **Collaboration:** Reports can be shared across teams easily.
- 5. **Predictive Insights:** Supports forecasting and trend analysis.

Conclusion:

Power BI empowers businesses to make faster, data-backed decisions and improves operational efficiency.

85. Compare Power BI with other visualization tools like Tableau or Excel.

Feature	Power BI	Tableau	Excel
Developer	Microsoft	Salesforce	Microsoft
Ease of Use	Very user-friendly	Slightly technical	Very easy
Data Handling	Good for medium to large data	Excellent for large data	Limited for big data
Integration	Integrates well with MS tools (Excel, Azure)	Integrates with multiple platforms	Works best for small datasets
Cost	Lower and affordable	Higher cost	Included with MS Office
Best For	Business analytics and reporting	Advanced data visualization	Basic analysis and charts

Conclusion:

Power BI offers a balance between **ease, power, and cost**, making it a top choice for most business users.

Integrated and Analytical Questions — 2-Mark Answers

86. Define data-driven decision making.

Answer:

Data-driven decision making (DDDM) is the process of using **data analysis and insights** to guide business strategies and actions instead of relying on intuition or guesswork.

87. What is the relationship between data analytics and business intelligence?

Answer:

Data analytics and business intelligence (BI) are closely related —

- **Business Intelligence** focuses on **descriptive analysis** (what happened).
- **Data Analytics** includes **predictive and prescriptive analysis** (why and what will happen).

Together, they help organizations make informed and strategic decisions.

88. What is big data analytics?

Answer:

Big data analytics refers to the process of **examining large and complex datasets** (structured or unstructured) to uncover hidden patterns, correlations, and trends for better decision-making.

89. What are KPIs (Key Performance Indicators)?

Answer:

KPIs are **quantifiable metrics** used to measure how effectively an organization is achieving its key business objectives.

Example: Sales growth rate, customer retention rate, and profit margin.

90. What is data cleaning?

Answer:

Data cleaning is the process of **detecting and correcting errors or inconsistencies** in data to improve its quality and accuracy before analysis.

5-Mark Questions & Answers

91. Explain the life cycle of data analytics.

Answer:

The **Data Analytics Life Cycle** describes the steps involved in extracting insights from data.

Phases:

1. **Data Collection:** Gathering data from various sources such as files, databases, or APIs.
2. **Data Cleaning:** Removing errors, duplicates, and inconsistencies.
3. **Data Exploration:** Understanding data patterns and distributions.
4. **Data Modeling:** Applying statistical or machine learning models to analyze data.
5. **Interpretation:** Interpreting results and generating insights.
6. **Visualization & Reporting:** Presenting findings through charts or dashboards.

Conclusion:

This life cycle ensures that data is handled systematically to produce reliable and actionable results.

92. Discuss the challenges in data collection and preprocessing.

Answer:

Challenges in Data Collection:

1. **Data Quality Issues:** Missing or incorrect data.
2. **Integration Problems:** Combining data from multiple formats or sources.
3. **Data Volume:** Handling large amounts of information.
4. **Privacy Restrictions:** Limited access due to security policies.

Challenges in Data Preprocessing:

1. **Dealing with Missing Values.**
2. **Removing Duplicates and Noise.**
3. **Data Transformation Complexity.**
4. **Time and Resource Intensive Process.**

Conclusion:

Effective data preprocessing ensures high-quality input, which improves analysis accuracy.

93. Explain the importance of statistics in data analytics.**Answer:**

Statistics is the **foundation of data analytics**, helping convert raw data into meaningful insights.

Importance:

1. **Data Summarization:** Mean, median, and mode describe data concisely.
2. **Understanding Variability:** Variance and standard deviation show spread.
3. **Decision Making:** Hypothesis testing supports evidence-based decisions.
4. **Prediction:** Regression and probability models forecast future outcomes.
5. **Data Interpretation:** Helps identify patterns and trends.

Conclusion:

Without statistics, data analytics would lack accuracy and scientific reasoning.

94. Discuss the role of visualization in data-driven decision making.**Answer:**

Visualization plays a crucial role in turning data into understandable insights.

Role:

1. **Simplifies Complex Data:** Converts large datasets into clear visuals.
2. **Reveals Patterns:** Helps spot trends, correlations, and outliers.
3. **Improves Communication:** Makes reports understandable for all users.
4. **Enhances Decision Making:** Enables quick and confident decisions.
5. **Interactive Exploration:** Allows users to explore data dynamically (e.g., in Power BI).

Conclusion:

Visualization bridges the gap between raw data and business understanding.

95. Describe a real-life use case where data analytics improved organizational efficiency.

Example:

Company: Amazon

Use Case:

- Amazon uses **data analytics** to analyze customer behavior and recommend products.
- It applies **machine learning models** to personalize user experiences and optimize pricing.
- **Outcome:** Increased sales, improved customer satisfaction, and reduced delivery costs.

Conclusion:

Through analytics, organizations can make smarter, faster, and more efficient decisions.

96. Explain how Python and Power BI complement each other in analytics.

Aspect	Python	Power BI
Purpose	Used for data cleaning, statistical analysis, and modeling.	Used for data visualization and reporting.
Strength	Strong in automation, AI, and complex computations.	Strong in dashboards and visual insights.
Integration	Can be embedded in Power BI for custom visuals.	Can run Python scripts for advanced analysis.
Output	Produces processed or modeled data.	Presents final insights interactively.

Conclusion:

Python handles **data processing and modeling**, while Power BI handles **visualization and presentation**, making them a perfect analytics combination.

97. Write short notes on:

a) Data Preprocessing

It is the process of **preparing raw data** for analysis by handling missing values, removing duplicates, and normalizing formats to improve model performance.

b) Data Transformation

This involves **modifying data** into a suitable format, such as converting data types, merging tables, or scaling values for consistency.

c) Data Modeling

Data modeling defines **relationships between data tables** and applies mathematical or machine learning models to extract patterns and insights.

98. Describe the process of building an end-to-end data analytics project.

Steps:

1. **Define the Problem:** Understand the business question to be solved.
2. **Collect Data:** Gather data from internal or external sources.
3. **Preprocess Data:** Clean and prepare data for analysis.
4. **Analyze & Model:** Apply statistical or machine learning methods.
5. **Visualize Results:** Use tools like Power BI or Python to present findings.
6. **Deploy & Monitor:** Share reports and update them regularly.

Conclusion:

An end-to-end analytics project converts raw data into insights that guide strategic business actions.

99. Explain the ethical issues and privacy concerns in data analytics.

Ethical Issues:

1. **Data Misuse:** Using data beyond its intended purpose.
2. **Bias in Data:** Leads to unfair or discriminatory decisions.
3. **Lack of Transparency:** Hiding how data is used or processed.

Privacy Concerns:

1. **Unauthorized Access:** Breach of confidential data.
2. **Data Ownership:** Unclear who owns the data.

3. **User Consent:** Collecting data without user permission.

Conclusion:

Organizations must follow ethical standards and privacy laws like **GDPR** to ensure responsible data usage.

100. Summarize the complete workflow of data analytics from data collection to visualization.

Stage	Description
1. Data Collection	Gather raw data from sources such as databases, sensors, or web APIs.
2. Data Cleaning	Remove missing values, duplicates, and errors.
3. Data Transformation	Convert and standardize data for consistency.
4. Data Analysis/Modeling	Apply statistical or machine learning models.
5. Interpretation	Understand patterns and results.
6. Visualization	Present insights through charts, dashboards, or reports (e.g., in Power BI).
7. Decision Making	Use insights for planning, forecasting, and performance improvement.

Conclusion:

This workflow ensures that data is transformed into meaningful insights for effective business decisions.

Problem 1 —

In a classification problem, a logistic regression model gives the following probabilities for five observations: [0.8, 0.3, 0.6, 0.9, 0.2]. If the cutoff is 0.5, determine which observations are classified as positive and compute accuracy, given the actual labels are [1, 0, 1, 1, 0].

Given: probabilities = [0.8, 0.3, 0.6, 0.9, 0.2]

Cutoff = 0.5 → predict 1 if prob $\geq$ 0.5 else 0.

Actual labels = [1, 0, 1, 1, 0]

Predictions:

- 0.8 → 1
- 0.3 → 0
- 0.6 → 1
- 0.9 → 1
- 0.2 → 0

So predicted = [1, 0, 1, 1, 0]

Compare with actual: all 5 match.

Confusion-matrix counts (using positive = 1):

- True Positives (TP) = 3
- True Negatives (TN) = 2
- False Positives (FP) = 0
- False Negatives (FN) = 0

Accuracy = (TP + TN) / total = (3 + 2) / 5 = 5/5 = 1.0 = 100%

Problem 2 —

Perform one iteration of K-Means clustering for data points (2,4), (3,5), (10,8), (11,9) with initial centroids (2,4) and (10,8). Assign points to clusters and compute new centroids.

Data points: (2,4), (3,5), (10,8), (11,9)

Initial centroids: C1 = (2,4), C2 = (10,8)

Assign each point to nearest centroid (Euclidean distance):

- (2,4): dist to C1 = 0 ; to C2 = $\sqrt{((2-10)^2 + (4-8)^2)} = \sqrt{(64+16)} = \sqrt{80} \approx 8.944 \rightarrow$ **Cluster 1**
- (3,5): dist to C1 = $\sqrt{((3-2)^2 + (5-4)^2)} = \sqrt{(1+1)} = \sqrt{2} \approx 1.414$; to C2 $\approx \sqrt{58} \approx 7.616 \rightarrow$ **Cluster 1**
- (10,8): dist to C1 ≈ 8.944 ; to C2 = 0 $\rightarrow$ **Cluster 2**
- (11,9): dist to C1 ≈ 10.296 ; to C2 = $\sqrt{((11-10)^2 + (9-8)^2)} = \sqrt{2} \approx 1.414 \rightarrow$ **Cluster 2**

Clusters after assignment:

- Cluster 1: (2,4), (3,5)
- Cluster 2: (10,8), (11,9)

New centroids (mean of points in each cluster):

- New C1 = $((2 + 3)/2, (4 + 5)/2) = (2.5, 4.5)$
 - New C2 = $((10 + 11)/2, (8 + 9)/2) = (10.5, 8.5)$
-

Problem 3 —

In K-Means clustering, the coordinates of two cluster centers are (2,3) and (8,7). Find the Euclidean distance between them.

Compute difference per coordinate: $\Delta x = 8 - 2 = 6$, $\Delta y = 7 - 3 = 4$

Distance = $\sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{6^2 + 4^2} = \sqrt{36 + 16} = \sqrt{52} \approx \mathbf{7.2111}$

(You can also give it as $\sqrt{52}$ if an exact form is preferred.)

Q4. Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	50 (TP)	10 (FN)
Actual Negative	5 (FP)	35 (TN)

Calculate **Accuracy** and **Precision**.

Formulas

- **Accuracy** = $(TP + TN) / (TP + TN + FP + FN)$
 - **Precision** = $TP / (TP + FP)$
-

Substitute values

- TP = 50
- TN = 35
- FP = 5
- FN = 10

$$\begin{aligned}\text{Accuracy} &= (50 + 35) / (50 + 35 + 5 + 10) \\ &= 85 / 100 \\ &= \mathbf{0.85 = 85\%}\end{aligned}$$

$$\begin{aligned}\text{Precision} &= 50 / (50 + 5) \\ &= 50 / 55 \\ &= \mathbf{0.909 \approx 90.9\%}\end{aligned}$$

✅ **Final Answers:**

- **Accuracy = 85%**
- **Precision = 90.9%**

Q5. A company predicts sales (Y) based on advertising expenditure (X) using the regression line: $Y = 5 + 0.8X$. Find the predicted sales when advertising expenditure is ₹12,000. Also, find the residual if actual sales are ₹14,000.

Given regression line:

$$Y = 5 + 0.8X$$

When $X = ₹12,000$:

$$\begin{aligned}\hat{Y} &= 5 + 0.8(12,000) \\ &= 5 + 9,600 \\ &= \mathbf{₹9,605}\end{aligned}$$

(Predicted Sales = ₹9,605)

If actual sales = ₹14,000, then

$$\begin{aligned}\text{Residual} &= \text{Actual} - \text{Predicted} \\ &= 14,000 - 9,605 \\ &= \mathbf{₹4,395}\end{aligned}$$

✅ **Final Answers:**

- **Predicted Sales = ₹9,605**
- **Residual = ₹4,395**

6. Find the mean, median, and mode of the data set: 5, 7, 3, 7, 9, 7, 2, 10.

Sort the data: 2, 3, 5, 7, 7, 7, 9, 10

1. **Mean** = (sum of values)/n
Sum = $5+7+3+7+9+7+2+10 = 50$.
 $n = 8 \rightarrow \text{Mean} = 50 / 8 = \mathbf{6.25}$.

2. **Median** (n is even = 8) = average of 4th and 5th sorted values = $(7 + 7)/2 = 7$.
3. **Mode** = value occurring most often = **7** (appears 3 times).

Answers: Mean = **6.25**, Median = **7**, Mode = **7**.

7. Calculate the range and standard deviation for the following values: 12, 15, 10, 18, 20.

1. **Range** = max – min = $20 - 10 = 10$.
2. **Mean** = $(12+15+10+18+20) / 5 = 75 / 5 = 15$.
3. Compute squared deviations from mean:
 - $(12-15)^2 = (-3)^2 = 9$
 - $(15-15)^2 = 0^2 = 0$
 - $(10-15)^2 = (-5)^2 = 25$
 - $(18-15)^2 = 3^2 = 9$
 - $(20-15)^2 = 5^2 = 25$Sum of squares = $9 + 0 + 25 + 9 + 25 = 68$.
4. **Population variance** = $68 / 5 = 13.6$.
Population standard deviation = $\sqrt{13.6} \approx 3.688$ (rounded to 3 decimals).

Answers: Range = **10**, Standard deviation $\approx$ **3.688** (population).

8. company records the monthly sales (in ₹ thousands) as: 25, 28, 30, 32, 27, 26, 29, 31, 34, 30. Calculate the mean, variance, and standard deviation.

25, 28, 30, 32, 27, 26, 29, 31, 34, 30 ($n = 10$)

1. **Sum** = $25+28+30+32+27+26+29+31+34+30 = 292$.
Mean = $292 / 10 = 29.2$ (₹ thousands).
2. Deviations and squared deviations ($x_i - 29.2$):
 - $25 \rightarrow -4.2 \rightarrow 17.64$
 - $28 \rightarrow -1.2 \rightarrow 1.44$
 - $30 \rightarrow 0.8 \rightarrow 0.64$
 - $32 \rightarrow 2.8 \rightarrow 7.84$

- $27 \rightarrow -2.2 \rightarrow 4.84$
- $26 \rightarrow -3.2 \rightarrow 10.24$
- $29 \rightarrow -0.2 \rightarrow 0.04$
- $31 \rightarrow 1.8 \rightarrow 3.24$
- $34 \rightarrow 4.8 \rightarrow 23.04$
- $30 \rightarrow 0.8 \rightarrow 0.64$

Sum of squared deviations = **69.60**.

3. **Population variance** = $69.60 / 10 = \mathbf{6.96}$ (₹ thousands)²).

Population standard deviation = $\sqrt{6.96} \approx \mathbf{2.639}$ (₹ thousands).

Answers: Mean = **29.2** (₹ thousands), Variance = **6.96**, Std. Dev $\approx$ **2.639** (₹ thousands).

9. The marks of 10 students are given below: 25, 30, 40, 35, 20, 45, 50, 55, 60, 40.

Find the coefficient of variation and interpret the result.

25, 30, 40, 35, 20, 45, 50, 55, 60, 40

1. **Sum** = 400 $\rightarrow$ **Mean** = $400 / 10 = \mathbf{40}$.

2. Squared deviations:

- $25 \rightarrow (-15)^2 = 225$
- $30 \rightarrow (-10)^2 = 100$
- $40 \rightarrow 0^2 = 0$
- $35 \rightarrow (-5)^2 = 25$
- $20 \rightarrow (-20)^2 = 400$
- $45 \rightarrow 5^2 = 25$
- $50 \rightarrow 10^2 = 100$
- $55 \rightarrow 15^2 = 225$
- $60 \rightarrow 20^2 = 400$
- $40 \rightarrow 0^2 = 0$

Sum = **1500**.

3. **Population variance** = $1500 / 10 = \mathbf{150}$.

Population standard deviation = $\sqrt{150} \approx \mathbf{12.247}$.

4. **Coefficient of Variation (CV)** = (Std Dev / Mean) $\times$ 100% = (12.247 / 40) $\times$ 100% $\approx$ **30.62%**.

Interpretation: CV $\approx$ **30.62%** — the standard deviation is about 30.6% of the mean, indicating a **moderate-to-high relative variability** in marks.

10. A bag contains 5 red, 3 blue, and 2 green balls. One ball is drawn at random. Find the probability of drawing a red ball.

Bag contents: 5 red, 3 blue, 2 green $\rightarrow$ **total = 10** balls.

Probability(red) = number of red / total = 5 / 10 = **1/2 = 0.5 = 50%**

Q11. A discrete random variable X has the following probability distribution:

X	0	1	2	3	4
P(X)=	.1	.2	.3	.3	.1

Find **E(X)** (mean), **Variance(X)**, and **Standard Deviation(X)**.

1 Mean (Expected Value), E(X)

$$E(X) = \sum X \cdot P(X)$$

$$E(X) = (0)(0.1) + (1)(0.2) + (2)(0.3) + (3)(0.3) + (4)(0.1)$$

$$E(X) = 0 + 0.2 + 0.6 + 0.9 + 0.4 = 2.1$$

 **Mean = 2.1**

2 Variance, Var(X)

$$Var(X) = \sum [X^2 \cdot P(X)] - [E(X)]^2$$

Compute $\sum X^2 \cdot P(X)$:

X	P(X)	X ²	X ² ·P(X)
0	0.1	0	0
1	0.2	1	0.2
2	0.3	4	1.2

X	P(X)	X ²	X ² ·P(X)
3	0.3	9	2.7
4	0.1	16	1.6

$$\sum X^2 P(X) = 0 + 0.2 + 1.2 + 2.7 + 1.6 = 5.7$$

$$\text{Var}(X) = 5.7 - (2.1)^2 = 5.7 - 4.41 = 1.29$$

✓ Variance = 1.29

3 Standard Deviation

$$SD = \sqrt{\text{Var}(X)} = \sqrt{1.29} \approx 1.136$$

✓ Standard Deviation = 1.136

✓ Final Answers

Measure	Symbol	Value
Mean	E(X)	2.1
Variance	Var(X)	1.29
Standard Deviation	SD(X)	1.136